Worksheet 2.4 — Rational, Radical & Higher-Order Derivatives

Grade 12 Calculus and Vectors (MCV4U) · Unit 2: Derivative Rules

The Big Ideas

Rewrite as powers; differentiate again for f''; use implicit differentiation when y is tangled with x.

- Rewrite: $\dfrac{1}{x^k} = x^{-k}$, $\sqrt[n]{x^m} = x^{m/n}$.
- Second derivative $f'' = (f')'$.
- Implicit: differentiate both sides, attach $\frac{dy}{dx}$ to every y.

The Big Picture

Rewrite then differentiate

Rational and radical functions are easiest as powers: write $\dfrac{1}{x} = x^{-1}$ and $\sqrt{x} = x^{1/2}$, then apply the power rule.

Combine rules

These often need the chain, product, or quotient rule together identify the structure before differentiating.

Higher-order derivatives

Differentiating again gives $f''(x)$, the rate of change of the slope, and so on for further derivatives.

Worked Examples

Example 1 — Rational via powers

Differentiate $f(x) = \dfrac{1}{x^2}$.

Solution. **Step 1 (rewrite as a power):** $\dfrac{1}{x^2} = x^{-2}$.

Step 2 (apply the power rule): $-2x^{-3}$.

Step 3 (rewrite back as a fraction): $-2x^{-3} = -\dfrac{2}{x^3}$.

Conclusion: $f'(x) = -\dfrac{2}{x^3}$.

Example 2 — Radical via powers

Differentiate $f(x) = \sqrt[3]{x^2}$.

Solution. **Step 1 (rewrite as a power):** $\sqrt[3]{x^2} = x^{2/3}$.

Step 2 (apply the power rule): $\frac{2}{3}x^{2/3-1} = \frac{2}{3}x^{-1/3}$.

Step 3 (rewrite back as a radical): $x^{-1/3} = \dfrac{1}{\sqrt[3]{x}}$.

Conclusion: $f'(x) = \dfrac{2}{3\sqrt[3]{x}}$.

Example 3 — Second derivative

Find $f''(x)$ for $f(x) = x^4$.

Solution. **Step 1 (find f' with the power rule):** $f'(x) = 4x^3$.

Step 2 (differentiate f' again to get f''): $f''(x) = \dfrac{d}{dx}(4x^3) = 12x^2$.

Conclusion: $f''(x) = 12x^2$. For x^3: f, f', f'' are $x^3, 3x^2, 6x$ — each the slope function of the previous:



Example 4 — Second derivative

Find $f''(x)$ for $f(x) = x^3 - 3x$ and $f''(1)$.

Solution. **Step 1 (find f'):** $f'(x) = 3x^2 - 3$.

Step 2 (differentiate again to get f''): $f''(x) = 6x$.

Step 3 (substitute $x = 1$): $f''(1) = 6(1)$.

Conclusion: $f''(x) = 6x$, and $f''(1) = 6$.

Example 5 — Implicit differentiation

Find $\dfrac{dy}{dx}$ for $x^2 + y^2 = 25$.

Solution. **Step 1 (differentiate both sides with respect to x):** $\dfrac{d}{dx}(x^2) + \dfrac{d}{dx}(y^2) = \dfrac{d}{dx}(25)$.

Step 2 (use the chain rule on y^2, since y is a function of x): $2x + 2y\dfrac{dy}{dx} = 0$.

Step 3 (isolate $\frac{dy}{dx}$): $2y\dfrac{dy}{dx} = -2x$.

Conclusion: $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

Example 6 — Rational via powers

Differentiate $f(x) = \dfrac{1}{x^4}$.

Solution. **Step 1 (rewrite as a power):** $\dfrac{1}{x^4} = x^{-4}$.

Step 2 (apply the power rule): $-4x^{-5}$.

Step 3 (rewrite back as a fraction): $-4x^{-5} = -\dfrac{4}{x^5}$.

Conclusion: $f'(x) = -\dfrac{4}{x^5}$.

Example 7 — Radical via powers

Differentiate $f(x) = x^{3/2}$.

Solution. **Step 1 (identify n):** $n = \frac{3}{2}$ — already a power, no rewriting needed.

Step 2 (apply the power rule): $\frac{3}{2}x^{3/2-1} = \frac{3}{2}x^{1/2}$.

Conclusion: $f'(x) = \frac{3}{2}x^{1/2}$.

Example 8 — Second derivative

Find $f''(x)$ for $f(x) = 2x^3 - x^2$.

Solution. **Step 1 (find f'):** $f'(x) = 6x^2 - 2x$.

Step 2 (differentiate again to get f''): $f''(x) = 12x - 2$.

Conclusion: $f''(x) = 12x - 2$.

Example 9 — Implicit differentiation

Find $\dfrac{dy}{dx}$ for $xy = 4$.

Solution. **Step 1 (differentiate both sides; the left side needs the product rule, since it's x times $y(x)$):** $\dfrac{d}{dx}(xy) = (1)y + x\dfrac{dy}{dx}$.

Step 2 (right side): $\dfrac{d}{dx}(4) = 0$, so $y + x\dfrac{dy}{dx} = 0$.

Step 3 (isolate $\frac{dy}{dx}$): $x\dfrac{dy}{dx} = -y$.

Conclusion: $\dfrac{dy}{dx} = -\dfrac{y}{x}$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = \dfrac{1}{x^5}$.

Question 2

Differentiate $f(x) = \sqrt[3]{x}$.

Question 3

Find $f''(x)$ for $f(x) = x^5$.

Question 4

Find $f''(x)$ for $f(x) = x^4 - 2x^2$.

Question 5

Find $\dfrac{dy}{dx}$ for $x^2 + y^2 = 9$.

Question 6

Differentiate $f(x) = x^{5/2}$.

Question 7

Find $f''(1)$ for $f(x) = x^3$.

Question 8

Find $f''(x)$ for $f(x) = 3x^4$.

Question 9

Find $\frac{dy}{dx}$ for $x^2 - y^2 = 1$.

Question 10

Differentiate $f(x) = \dfrac{2}{x^3}$.

Question 11

Find $f''(x)$ for $f(x) = x^3 - 6x^2 + 2x$.

Question 12

Find $\dfrac{dy}{dx}$ for $xy = 10$.

Question 13 — Challenge

Find $\dfrac{dy}{dx}$ for $x^2 + xy + y^2 = 7$ (product rule on xy).

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = \dfrac{1}{x^5}$.
2. Differentiate $f(x) = \sqrt[3]{x}$.
3. Find $f''(x)$ for $f(x) = x^5$.
4. Find $f''(x)$ for $f(x) = x^4 - 2x^2$.
5. Find $\dfrac{dy}{dx}$ for $x^2 + y^2 = 9$.
6. Differentiate $f(x) = x^{5/2}$.
7. Find $f''(1)$ for $f(x) = x^3$.
8. Find $f''(x)$ for $f(x) = 3x^4$.
9. Find $\dfrac{dy}{dx}$ for $x^2 - y^2 = 1$.
10. Differentiate $f(x) = \dfrac{2}{x^3}$.
11. Find $f''(x)$ for $f(x) = x^3 - 6x^2 + 2x$.
12. Find $\dfrac{dy}{dx}$ for $xy = 10$.
13. Find $\dfrac{dy}{dx}$ for $x^2 + xy + y^2 = 7$ (product rule on xy).

✓ Answer Key

1. $-\frac{5}{x^6}$
2. $\frac{1}{3}x^{-2/3}$
3. $f'' = 20x^3$
4. $f'' = 12x^2 - 4$
5. $-\frac{x}{y}$
6. $\frac{5}{2}x^{3/2}$
7. $f'' = 6x \Rightarrow 6$
8. $f'' = 36x^2$
9. $\frac{x}{y}$
10. $-\frac{6}{x^4}$
11. $f'' = 6x - 12$
12. $-\frac{y}{x}$
13. $2x + y + xy' + 2yy' = 0 \Rightarrow y' = -\dfrac{2x + y}{x + 2y}$